

Evaluation Framework: Nine Properties over Distribution Labels, with a Formal
Stance Model

Why not a distribution label I

“Secure Linux” is not a property of a distribution; it is an outcome of a deployment matching a threat model. Distribution labels hide exactly the heterogeneity that matters: one candidate may pair a strong containment architecture with a weak update-operations story, another may have excellent supply-chain discipline and no runtime confinement at all. A single label collapses those differences into a ranking question that no available evidence answers — and ranking invites the numeric composite scores (“Qubes 9.4” style) that this review refuses, for reasons given below. The alternative, inherited from the source assessment, is to evaluate every candidate against a fixed set of properties, each phrased as a question a deployment can be interrogated with.

The nine properties I

tbl. 1 states the 9 properties used throughout the review, with the question each property asks and its security significance in this threat model.

The nine properties I

Table 1: The 9 evaluation properties: the question each property asks and its security significance in this threat model.

Property	Question to ask	Security significance
Containment	What remains protected after the browser or agent is fully controlled?	A prevention failure should not automatically become whole-machine compromise.
Authority	What can the process already read, transmit, sign, or change?	Excessive legitimate permissions can bypass the need for an exploit; this is the authorized-misuse axis from sec. 3.
Trusted computing base	Which privileged components must all remain correct?	A small, carefully constrained boundary is preferable to many privileged integrations; it is also what patch operations must cover (Saltzer and Schroeder 1975).
Application confinement	Are permissions restrictive by default and actually compatible with the applications?	A theoretical policy is not protection if users routinely disable it.
Integrity	What verifies the boot chain and deployed software, and who holds the keys?	Authenticity, runtime integrity, and resistance to rollback are distinct properties, not one checkbox.
Persistence and recovery	Which state survives replacement or reboot?	Rebuilding the system is insufficient if malicious user state or stolen credentials survive.
Update operations	How quickly do fixes reach every active environment?	A strong architecture with neglected templates or pinned dependencies can lose its advantage — the QSB-118 lesson in sec. 5.
Supply-chain trust	Who may introduce or approve new executable code?	Reproducibility and signatures answer different questions from whether code is benign (N. Project 2026).

The nine properties II

Human usability

Will the owner preserve the intended boundaries during real work?

Sustainable security is more valuable than a configuration abandoned under pressure.

The nine properties I

The properties are evaluation criteria, not a scoring model. They overlap deliberately where the underlying security properties genuinely interact (authority with containment, trusted computing base with update operations), and the overlap is visible in the analysis rather than averaged away.

Parallel threat-modeling structures, and why a property matrix still wins for OS work I

The property matrix is not the only structured lens for agentic systems, and the 2025–2026 standardization wave produced alternatives worth naming. OWASP's Agentic AI Threats and Mitigations guide catalogs threat scenarios for agent systems — memory manipulation, tool misuse, identity and privilege confusion — and its December 2025 Top 10 for Agentic Applications distills them into a ranked practitioner list (Foundation 2025a, 2025b). CSA's MAESTRO frames agentic risk as seven interdependent layers spanning the agent's reasoning, tools, and multi-agent interconnects (Alliance 2025), and its swarm governance work extends the framing with zero-trust patterns such as short-lived agent identities (Alliance 2026). These structures earn their place: they enumerate agent-stack threats with a granularity the nine properties

Parallel threat-modeling structures, and why a property matrix still wins for OS work I

deliberately do not attempt, and the synthesis chapters borrow from them where agent-internal surfaces dominate (sec. 11; sec. 13).

For operating-system evaluation, though, the property matrix is the better lens. First, the alternatives are keyed to the agent stack — memory, tools, orchestration layers — where an OS review needs the boundary keyed to what a kernel or hypervisor can actually enforce: containment, authority, and the trust base that patches must cover. The 2026 incident record supports the distinction: the AISI and OpenAI–Hugging Face incidents were bounded by sandboxes, connectivity policy, and revocation — OS-level mechanisms — while the persuasion of a human maintainer traveled through an entirely different channel (Institute 2026; OpenAI 2026). A taxonomy that does not force that separation cannot rank platforms. Second, the frameworks are threat lists, not evaluation instruments: they say what can go wrong, not what

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a candidate's documented design provides, whereas the nine properties convert to per-candidate stances that can be diffed, tested, and refreshed as advisories land (sec. 5; sec. 6). Third, they abstract away the trusted computing base — which the 2026 advisory record (QSB-118, the Nix advisories) shows to be decisive under automation pressure.

The evidence discipline is the same one the threat model applies to its baseline. The NCSC's original AI cyber-threat assessment (January 2024) and its "From Now to 2027" successor (May 2025) supply the capability horizon this framework evaluates against (Centre 2024, 2026) — faster exploitation of known vulnerability classes, not assumed universal autonomous compromise — rather than any single vendor's capability narrative.

The anti-scoring stance I

There is no adequate evidence in this review for assigning meaningful universal scores, and raw vulnerability counts would not resolve the comparison either: they conflate disclosure volume with exposure, and they are confounded by project size, patch latency, and measurement effort. Worse, composite scores create false precision across incommensurable properties — trading a point of “usability” against a point of “containment” implies an exchange rate that no operator’s actual threat model supplies. This is the same discipline that separates reproducible deployment from verified reproducible builds (N. Project 2026) and authenticity from provenance: properties that answer different questions must stay separate.

The anti-scoring stance I

Instead, every candidate–property cell receives one qualitative stance: **strong**, **partial**, **weak**, or **n_a**. A **strong** stance means the documented design directly and coherently addresses the property in this threat model. **partial** means the property is addressed with material conditions, integration gaps, or operator obligations attached. **weak** means the documented design provides little of what the property asks. **n_a** marks candidates for which the property is out of scope or unverified by available evidence — used rather than guessed, because absence of a confirmed feature in this review is not proof the feature is unavailable.

A formal stance model I

The vocabulary is small enough to state formally, and the formal statement is worth writing down because it fixes what the matrix claims and what it refuses to claim. Let C be the set of 24 candidates, P the set of 9 properties, and $S = \{\textit{strong}, \textit{partial}, \textit{weak}\}$ the stance vocabulary.

The evaluation matrix is a total function: every candidate–property pair receives exactly one qualitative stance, and n/a marks the pairs a candidate does not engage at its architectural level:

$$\sigma: C \times P \rightarrow S \cup \{n/a\} \tag{1}$$

A formal stance model I

Read as a function, (**def:stance_mapping?**) makes two commitments that prose leaves implicit. It is **total**: the pipeline refuses to emit figures until all 216 cells are defined, so no candidate escapes evaluation by omission. And it is **qualitative**: the codomain carries no numbers, so the matrix cannot be averaged or ranked without an explicit modeling step — the step the anti-scoring stance above declines to supply.

The stance words themselves carry a preference structure rather than a measurement scale, and the review reads them as ordered:

A formal stance model I

Stances form a preference order over postures: a candidate whose documented design earns the left-hand stance dominates a candidate that earns the right-hand stance *on that property*:

$$strong \succ partial \succ weak \quad (2)$$

By (**def:stance_order?**) the order fixes postures, not distances between them: no arithmetic over stances is meaningful, and no rung is a probability. Notably, n_a does not appear in eq. 2 at all — an unverified or out-of-scope cell sits outside the preference order entirely, which is what stops “we could not verify” from masquerading as “we verified it is weak.” The formal model thus encodes exactly the discipline the vocabulary was built for: total coverage, ordered confidence in documented designs, and honesty about the unverified.

A formal stance model I

The stance vocabulary is ordinal without interval structure: strong and partial can be told apart, but no measurement assigns them distances, and an average over stances therefore implies a measurement that does not exist. By (**def:stance_order?**) the preference order is qualitative only — it fixes which posture dominates on a property, never by how much — so any composite score built on it would be arithmetic performed on words.

How the matrix is constructed and refreshed I

The matrix lives as data, not prose. In `src/agentos_security/registry.py`, each candidate carries a `property_stance` dictionary keyed by all 9 property identifiers, alongside a design summary, a named limitation, and an assessment. The pipeline emits the full 24-candidate $\times$ 9-property table to `output/data/evaluation_matrix.csv` — one row per pair — and self-checks enforce stance-vocabulary validity and matrix completeness before any figure is regenerated. Stances are assigned from documented designs, official security documentation, project advisories, and primary incident reports; vendor feature lists support `partial` or `strong` stances only where the project itself documents the mechanism, never as measured resistance results.

How the matrix is constructed and refreshed I

Refresh is a deliberate operation: a stance changes when the underlying documentation, an advisory, or a release note changes — for example, a fixed advisory can move an update-operations stance, while a new integration gap documented by the project can move an application-confinement stance. Because the matrix is deterministic data with the review date 2026-09-10 stamped into the build, a future refresh produces a diffable artifact: added rows, changed stances, and the reasons, rather than rewritten narrative. fig. 1 renders the current matrix.

The defensive stack: mitigation classes as a second lens I

The candidate–property matrix answers “how does each platform score on the evaluation properties?” — but an operator choosing *what combination of mechanisms* to deploy also needs the complementary question: which mitigation classes does each candidate actually ship? Agent-execution infrastructure now standardizes on a recognizable stack of kernel primitives — sandboxing namespaces, restricted user namespaces, seccomp filters, egress proxies — while microvisor and unikernel candidates push different classes (verified boot, disposable execution) to their limits (sec. 8). The defensive-stack matrix captures that second view as data, and treats layered controls as formal objects rather than a checklist: the Cognitive Integrity Framework’s Defense Composition Algebra reasons about defenses composed

The defensive stack: mitigation classes as a second lens I

over bounded-trust delegates, prior art this stack operationalizes as concrete per-class stances (Friedman 2026).

In `registry.py`, `MITIGATION_CLASSES` defines the eight mitigation classes — `memory_safety` (memory-safe implementation languages), `allocator_hardening` (hardened allocators and exploit mitigations), `sandboxing_primitives` (namespaces, `seccomp`, capability confinement), `mac_framework` (MAC frameworks), `verified_boot` (authenticated boot chains), `reproducible_deployment` (declarative, rebuildable state), `disposable_execution` (ephemeral task environments), and `update_automation` (prompt patch delivery). `DEFENSIVE_STACK` assigns every 24 candidate a stance in each class, using the same `strong/partial/weak/n_a` vocabulary as the property matrix, grounded in the candidates' documented features

The defensive stack: mitigation classes as a second lens I

and the 2026 research record (sandboxing stances reflect documented namespace and seccomp practice; verified-boot stances reflect measured-boot and Secure Boot implementations). The helper `defensive_stack_rows()` emits the full 24-candidate $\times$ 8-class table — 192 stance cells — and the analysis pipeline writes it to `output/data/defensive_stack.csv`, where the same self-checks that police the property matrix enforce vocabulary validity and completeness. fig. 2 renders the matrix as a heatmap with the eight candidate categories grouped and a per-class coverage marginal.

The defensive stack: mitigation classes as a second lens I

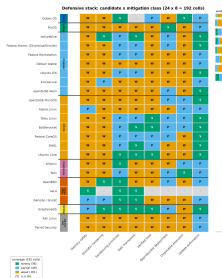


Figure 2: Defensive-stack coverage across eight mitigation classes: weak cells cluster on conventional desktops while compartmentalized, server, and high-assurance candidates concentrate strong stances in sandboxing, verified boot, and update operations.

The defensive stack: mitigation classes as a second lens I

Reading the two matrices together is the point. A candidate can hold a strong containment stance (Qubes, from domain separation) while its memory-safety stance reflects the mixed language base of its tooling; a reproducibility-first candidate (NixOS) holds strong reproducible-deployment and update-automation stances while its sandboxing and MAC stances remain weaker (sec. 6); a microvisor candidate may score strong on verified boot and disposable execution while offering no MAC framework at all (sec. 8). Neither matrix substitutes for the other: the property matrix carries the threat-model judgments, the defensive stack names the mechanism inventory an operator composes from. Both are deterministic artifacts stamped with the review date 2026-09-10, and both refresh by diffing rows, not rewriting adjectives.

What the matrix buys I

Three things. First, **comparability without false precision**: two candidates can be contrasted property by property, and the contrast shows *where* they differ (Qubes holds containment strongly and application confinement partially; NixOS holds update operations and supply-chain trust strongly and application confinement weakly) rather than by how much in aggregate. Second, **auditability**: each stance is a claim about documentation that a reader can check against the cited source, and the structural invariants are test-enforced — every candidate covers every property, every stance is in the vocabulary, and the matrix has 24×9 cells exactly. Third, **scenario grounding**: the per-scenario recommendations in sec. 16 select candidates by which properties the scenario's threat model weights most, which is precisely what a

What the matrix buys I

label-based ranking cannot do.

Limitations of matrix thinking I

A matrix is a discipline for judgment, not a substitute for it, and four limits should be kept in view.

Limitations of matrix thinking I

- ▶ **Stances are not measurements.** Every cell is an analytical judgment from documented designs (Saltzer and Schroeder 1975); none is a penetration-test result. The vocabulary encodes confidence in a documented capability, not a measured resistance rate against a live adversary. The defensive-stack stances inherit the same limit in sharper form: a strong sandboxing stance says the project documents and ships the mechanism, not that the mechanism holds against a named attacker.
- ▶ **Properties interact.** A strong containment stance can be nullified by a weak update-operations stance if the containment mechanism itself ships unpatched; a strong integrity stance does not constrain what an authorized agent does with verified software. Cell-level reading without row-level synthesis will mislead, which is why the scenario chapters, not the matrix, carry the recommendations.
- ▶ **Unverified defaults are a standing gap.** Where this review could not verify a current default or security property, the cell says so; a future audit may legitimately move stances in either direction, and the diffability of the matrix is designed for exactly that.

Limitations of matrix thinking II

- ▶ **The matrix is frozen in time at 2026-09-10.** Projects ship; advisories land; trajectories noted per candidate (for instance, the GUI-domain split in sec. 5 and boot-integrity work in sec. 6) may convert into stronger stances at the next refresh. Low-confidence forecasting of those movements is handled in sec. 15 rather than smuggled into current stances.

Limitations of matrix thinking I

With the framework fixed, the next two sections apply it in depth to the two anchor platforms whose properties the rest of the architecture inherits: Qubes OS for containment (Q. O. Project 2026) and NixOS for reproducible operations (community 2026).